

Prefab Configuration Guide

1) Player Prefab (`Player.prefab`)

Components:

- `SpriteRenderer`
- `Collider2D` (`BoxCollider2D` recommended)
- `Rigidbody2D` (optional; if used, set Gravity Scale = 0)
- `PlayerController`

Setup:

- Tag: `Player`
- Create child object `FirePoint` at ship nose (e.g., y = +0.6)
- Assign:
 - `Player Bullet Prefab`
 - `Fire Point`

2) Enemy Prefabs

A) Chaser Enemy (`Enemy_Chaser.prefab`)

Components:

- `SpriteRenderer`
- `Collider2D`
- `EnemyController`

EnemyController settings:

- Type: `Chaser`
- BaseHealth: `1`
- MoveSpeed: `2.8`
- ScoreValue: `100`

B) ZigZag Enemy (`Enemy_ZigZag.prefab`)

Components:

- `SpriteRenderer`
- `Collider2D`
- `EnemyController`

EnemyController settings:

- Type: `ZigZag`
- BaseHealth: `1`
- MoveSpeed: `2.3`
- ZigZagAmplitude: `1.5`
- ZigZagFrequency: `2.5`
- ScoreValue: `140`

C) Shooter Enemy (`Enemy_Shooter.prefab`)

Components:

- `SpriteRenderer`
- `Collider2D`
- `EnemyController`

Child object:

- `FirePoint` at lower nose (e.g., y = -0.6)

EnemyController settings:

- Type: `Shooter`
- BaseHealth: `2`
- MoveSpeed: `1.9`
- ShootInterval: `1.3`
- EnemyBulletPrefab: assign enemy bullet prefab
- FirePoint: assign child transform
- ScoreValue: `200`

3) Bullet Prefabs

A) Player Bullet (`Bullet_Player.prefab`)

Components:

- `SpriteRenderer`
- `Collider2D` (trigger ON)
- `BulletController`

BulletController settings:

- Owner: `Player`
- Speed: `12`
- Damage: `1`
- LifeTime: `4`

B) Enemy Bullet (`Bullet_Enemy.prefab`)

Components:

- `SpriteRenderer`
- `Collider2D` (trigger ON)
- `BulletController`

BulletController settings:

- Owner: `Enemy`
- Speed: `8`
- Damage: `1`
- LifeTime: `5`

4) Power-Up Prefabs

Common components:

- `SpriteRenderer`
- `Collider2D` (trigger ON)
- `PowerUpController`

A) PowerUp_RapidFire.prefab

- Type: RapidFire
- MoveSpeed: 2

B) PowerUp_Shield.prefab

- Type: Shield
- MoveSpeed: 2

C) PowerUp_Health.prefab

- Type: Health
- MoveSpeed: 2

5) Manager Object Assignments

SpawnManager

Assign references:

- Chaser Enemy Prefab
- ZigZag Enemy Prefab
- Shooter Enemy Prefab
- RapidFire PowerUp Prefab
- Shield PowerUp Prefab
- Health PowerUp Prefab

PlayerController

Assign references:

- Player Bullet Prefab
- Fire Point

Enemy Shooter Prefab

Assign:

- Enemy Bullet Prefab
- Fire Point